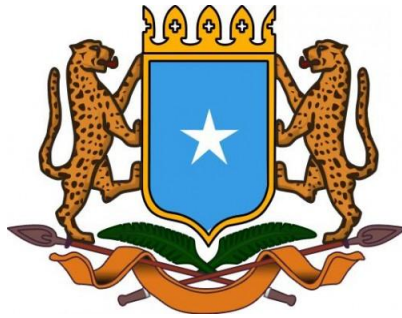




SOMALI NATIONAL INFRASTRUCTURE STRATEGY (SNIS)

2019-2063

Table of contents

1. Background	3
2. Introduction	4
3. Vision and overall objective for Somalia Infrastructure Development:	5
3.1 Overall Objective for the infrastructure sector in 2019 to 2023:	5
3.2 Vision.....	5
3.3 Mission	5
4. Situation Analysis of the Infrastructure Sector.....	6
4.1 Land and urban planning	6
4.2 Housing	7
4.3 Transport (Roads, Ports and Airports)	8
4.4 Development and Management of Water resources and Waste infrastructure	13
4.5 Energy	14
4.6 Information Communication and Technology (ICT).....	15
5. The <i>Main Challenges</i> hindering the Infrastructure Sector growth and expansion.....	17
6. Infrastructure Sector's <i>Development Objectives</i>	18
7. The Infrastructure Sector <i>Strategic Policies</i>	19
7.1 National Urban/ Land Policy	19
7.2 National Housing Policy	19
7.3 National Transport Policies	19
7.4 Sustainable Energy Investment Policy and Energy Strategy	20
7.5 Waste Management and Sanitation infrastructure Policy	20
7.6 ICT Policy	20
8. The Infrastructure Sector <i>Development Priorities</i>	21
9. The Main Infrastructure Sector's <i>Programs during the Plan Period 2019-2023</i>	22
10. Aggregated Resources Requirement	25
11. Milestone of <i>Infrastructure Sector</i> development (2019-2023).....	26
12. Coordination aspects and requirements with other Sectors/activities/regions	26
13. Conclusion.....	27
14. Intervention Programmes.....	28

1. Background

Infrastructure is subjected to damage and destruction during the course of any conflict with resultant decreased per capita access. As a result, restoration of infrastructure can be a major dividend of peace and is a critical factor in the success of post-conflict recovery. In conflict-sensitive environments, the effectiveness of infrastructure is a contributor to and a barometer to determine whether a society will slip back into violence or make a peaceful transition out of the conflict cycle.

Infrastructure encompasses institutions and human capabilities, as well as material and physical processes, systems and structures. Thus, restoration of infrastructure systems must include economic (transport, energy and communication) social infrastructure (water, sanitation, housing, schools and hospitals) and institutional infrastructure including the facilities, equipment, and personnel required for governance at the local and national levels. A vision and framework for planning, reconstruction and development of such infrastructure systems requires national and local in-depth stakeholder consultation to achieve legitimacy and ownership.

Infrastructure systems in its broadest sense can be described by three main constituent components: assets, knowledge and institutional capacity. Infrastructure systems are acting as the vehicle needed to deliver goods and services to ensure the effective functioning of society. Without any of the above three elements there will be neither a functional asset nor capacity (knowledge and institutions) to manage (develop, own, operate or maintain) the assets. Infrastructure systems can be described as the enabler through which delivery of goods and essential services to the economy and society as a whole are provided. The inability for infrastructure to perform effectively as the enabler to supply goods and services not only limits the economic development of a country or region but can stifle recovery and reconstruction in a post conflict environment.

Infrastructure development does not stop with the provision of built facilities; service provision that actually makes the physical infrastructure accessible to the citizens is likewise important. Roads are rehabilitated to provide access to markets and employment, hospitals are built to provide health services, government buildings are rehabilitated to provide the preconditions for good governance and service delivery and housing, water and sanitation are needed to provide adequate living conditions to IDPs, returnees, urban poor and other vulnerable groups. Demands placed upon infrastructure systems resulting from protracted displacement, migration and rapid urban growth have been neglected for decades and strained existing physical infrastructure to and beyond their functional limits.

In the case of Somalia, several consultations with stakeholders ranging from civil society, national and local authorities and international development partners have been undertaken, including assessments of the current state of infrastructure, the priority needs for reconstruction

and practical requirements to achieve priority improvements in the country. It is on this basis that this National Infrastructure Development Plan has been developed.

National infrastructure planning in Somalia involves at least five key federal ministries

- Ministry of Public Works, Housing and Reconstruction (incl. roads authority)
- Ministry of Energy & Water Resources
- Ministry of Transport & Civil Aviation
- Ministry of Ports & Marine Transport
- Ministry of Post, Telecommunication and technology
- Directorate of Environment the Office of the Prime Minister

2. Introduction

In line with the Federal Government of Somalia's endorsement, the Sustainable Development Goals (SDGs) will act as a guiding framework for the infrastructure sector development priorities with due consideration given to localizing the SDGs with the needs of the Somali people.

The consequences of over two decades of a civil war, that has had adverse effects on the political and economic systems of the country and culminated in complete institutional collapse which can be directly felt by many Somalis who as a result have no access to fundamental public services. Larger scale rehabilitation and expansion of infrastructure systems will have enormous impact on the perception and improvement of stability in the country.

The lack of an effective federal government since 1991 has significantly hindered the country's development in particular with regard to maintaining and expanding the public infrastructure and developing the necessary regulatory and governance frameworks. This has left Somalia faced with the complex and pressing challenges of state building and reconstruction in the absence of endogenous sources of public revenue, and with hardly any institutional capacity.

Development and rehabilitation of transport infrastructure (roads, ports and airports) will help re-establish the physical links destroyed by war, facilitate the strengthening of social links, and expand regional trade. In the road sub-sector, there is need to re-establish regional interconnectivity with countries such as Ethiopia, Kenya and Djibouti, and between the various regions within Somalia with the aim of supporting efforts towards peace and reconciliation and regional linkage. Agriculture and livestock account for 65% of the Somali economy, and employ around the same percentage of the Somali labor force. This value chain requires specific attention from the perspective of trade, transport and development corridors when planning the transport sector in the country, also with regard to exporting livestock through Somali ports.

Infrastructure to support the social sector, such as health facilities, schools, housing, water, sanitation and waste management are critical to provide decent living conditions for the Somali people. Somalia has some of the poorest indicators in health and education in the world, and it is essential to improve these types of infrastructure to allow the relevant social services actors to be able to effectively address these issues. The economic return from these type of investments will materialize in the form of improvements in health conditions and education attainment. There are over 1.1 million IDPs within Somalia and more than half a million Somali refugees in neighboring countries. All of these people will need housing, water, sanitation, health and education infrastructure if they are to willingly return or resettle in a sustainable and peaceful manner.

This chapter addresses some of these key challenges beginning with capturing both the short and long term vision of the Federal Infrastructure Ministries, then progressing on to analyze the present conditions of the infrastructure sector, the main challenges facing the infrastructure sector's growth and expansion, highlighting the sector's development objectives, strategic policies and sector development priorities. This chapter will also highlight the infrastructure sector's main programs needed to achieve the defined sectorial objectives for 2017 to 2019 while also assessing the aggregated resources required and planning the milestones for infrastructure development. Finally, it will engage in a discussion on the coordination mechanisms required to successfully implement the strategic objectives and activities across Somalia.

3. Vision and overall objective for Somalia Infrastructure Development:

3.1 Overall Objective for the infrastructure sector in 2019 to 2023:

Revitalize, operate and maintain essential infrastructure and services to improve stability and set foundations for social, economic and sustainable development.

3.2 Vision

Somalia created infrastructure that enhances the employment, increases food security, builds up resilience to climate change and variability, respect Somali cultural heritage and is environmentally and economically sustainable.

3.3 Mission

The short term (5 years) vision for Somalia's infrastructure is to lay out the foundations of resilient infrastructure systems¹ that will contribute to the countries stability and will provide the

¹ Resilience is the ability of assets, networks and systems to anticipate, absorb, adapt to and/ or rapidly recover from a disruptive event. Source: UK cabinet office, 'Section A: Introduction, Definitions and

necessary services for all (setup the organizational structures & capacity, infrastructure standards, legislations, priorities, strategies, mobilize resources and planning for long term infrastructure development, land and urban planning), while simultaneously rehabilitating, (re)constructing, upgrading, mapping and developing the existing infrastructure systems in compliance with SDGs, IGAD Infrastructure Master Plan and Africa Infrastructure Strategy to create infrastructure that enhances the employment, increases food security, builds up resilience to climate change and variability, respect Somali cultural heritages and is environmentally and economically sustainable.

In the long term (15 years/2030) Somalia is envisioning to have its infrastructure serve Somalia, the people of the Horn of Africa and IGAD countries, to facilitate economic and social development (interconnecting the nation and the region, with modern telecommunication, so that Somali citizens have access to affordable energy, water and appropriate waste disposal systems, adequate and sustainable housing, under the guidance of proper urban and territorial planning and financed and maintained in a sustainable manner.

Sustainability of existing and new or rehabilitated infrastructure must also be considered throughout the plan. Budget for operation and maintenance is essential to ensure the assets are kept in working condition and future repair costs are minimized.

4. Situation Analysis of the Infrastructure Sector

The collapse of the Somali State during the civil war culminated in the destruction of public and private infrastructure assets and systems including roads, seaports, airports, water, energy, sewage systems factories, hospitals, laboratories, schools, businesses and ministries as well as the loss of legislative and other key documents that provided regulatory or planning frameworks for the infrastructure sector. Protracted displacement, internal migration and rapid urbanization pose an additional challenge to service delivery and planning in the infrastructure sector. This impact can be felt when analyzing the present conditions of key infrastructure components.

4.1 Land and urban planning

The urbanization process in Somalia has been happening haphazardly, with no spatial planning or urban management. While some urban centers of the main settlement are relatively structured in terms of lay-out, based on pre-civil war planning attempts, much of the urban space, including vacant space within the formal urban fabric, holds pockets of ‘nomadic’ tents (‘buuls’) belonging

Principles of Infrastructure Resilience’,
https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/78902/section-a-natural-hazards-infrastructure.pdf

to internally displaced communities or recent arrivals of refugee returnees. As such, the impact of IDPs on land ownership and thus any type of intervention in urban areas is substantial. In some areas former farmland is being converted into ‘urban’ land by tracing a grid with a bulldozer, subdividing it into plots, leaving no space for public infrastructure and taking little account of potential natural hazards, for instance from flooding. IDP settlements are often concentrated in peri-urban areas of large cities and constitute there a significant settlement typology² which has grown uncontrolled and unplanned and is characterized by high density, lack of social services, absence of livelihood opportunities and devastating living conditions.

While the Somali construction sector is booming due to the current reconstruction activities in the country, the infrastructure ministries at federal and state level are operating in a grey zone without any up to date urban legislation. The majority of legislation for the infrastructure sector dates to a pre-war period and its status is either not clarified or not existing. Regional and local governments lack guidance through policies, norms and standards and the few current infrastructure rehabilitation projects are not built hand in hand with planning of related service provision.

The same applies to land management. With the exception of Puntland, land legislation on managing land and resolving land disputes at the municipal level is largely outdated.³

4.2 Housing

Access to adequate housing⁴ is a challenge growing fast with Somalia’s high urbanization rate⁵. 1,106,751 or 9% of the population or 14.7% of households are IDPs⁶ of which most live in camps and informal settlements. An increasing number of urban dwellers, especially the most poor and vulnerable groups (not only IDPs, but also women-headed households, refugee returnees, persons with disabilities and youth) are living in precarious conditions, addressing their housing needs informally, lacking access to basic services, isolated from livelihood opportunities and vulnerable to forced evictions or homelessness. Somalia’s public and private

² For instance, in Mogadishu 33% of the cities IDP live in the peri-urban district of Deynille in 142 informal settlements, 19% in Kaxda in 112 informal settlements and another 17% in Hodan. Source: UNHCR, ‘Internal displacement profiling in Mogadishu’, April 2016

³ Law N°10, 17 Dec. 1980 – Administration of Building Land for Banaadir Region, Law N°41, 13 Sep. 1973 – Urban Land Distribution, Decree 31, 31 Dec. 1973 – Nomination of Chairpersons of Regions and Districts, Law N° 73, 21 Oct. 1975 – Agricultural Land Law, Law N° 23, 16 Oct. 1976 – Law 23 on Agriculture. All of these laws are not valid as they do not relate to the current Constitution.

⁴ For housing to be adequate, it must, at a minimum, meet the following criteria: Security of tenure; Availability of services, materials, facilities and infrastructure; Affordability; Habitability; Accessibility; Location and Cultural Adequacy. Source: UNHCR and UN-Habitat, ‘The Right to Adequate Housing’, Fact Sheet No. 21 (Rev1)

⁵ 4.06% annual rate of urbanization growth (2010-15 est.), source CIA ‘The World Fact Book’. Mogadishu is the 2nd fastest growing city in the world.

⁶ Population Estimation Survey, Federal Government of Somalia and UNFPA, Oct 2014

housing stock has after more than 20 years of conflict been largely destroyed⁷. Adequate housing is not affordable to IDPs and the urban poor, and affordable housing is not adequate.

Protracted internal displacement situations exist in all regions, most of them in urban areas. The majority of IDPs settle in informal and unplanned settlements with inadequate housing, regularly at risk of forced eviction and in deplorable and impoverished living conditions. Most settlements are controlled and dominated by gatekeepers. The urban dimension of the protracted nature of displacement is evident and in many instances IDP settlements have turned into urban slums and sites of urban impoverishment, where economic migrants, returnees, refugees and urban poor have joined IDPs. While protracted displacement is a rather static situation, newly displaced persons regularly join existing IDP settlements adding further pressure on the community. Furthermore, IDPs are not a homogenous group when it comes to their displacement history. Many have been displaced during different phases and types of armed conflict, others by clan conflict, by drought or famine or by other disasters triggered by natural hazards, such as recurrent floods. Still others had to move due to forced evictions. Many IDPs have been repeatedly displaced several times due to various causes.⁸

The Somali population is growing by almost 3% per year, with a high fertility rate of 6.26 children per woman, which is the 4th highest in the world.⁹ Based on this anticipated growth rate the Somali population will grow to approx. 14.3M people by the end of 2019 or by 1.2M people in the three year period covered by the NDP. Housing strategies need therefore take into account not only the deficit of adequate housing for the current population, in particular for the urban poor, IDPs and refugee returnees but also plan for the natural population growth.

4.3 Transport (Roads, Ports and Airports)¹⁰

Transport vision Efficient transport infrastructure is essential within an economy as it dictates the type of economic and social activities that can be conducted within a particular section of the country. One of the most important factors for making investment decisions is the availability of proper transport networks that facilitate the movement of goods, services as well as workers. Moreover, a well-developed transport network enables the underdeveloped communities to access essential goods and services. The transport infrastructure is also particularly important in enabling connectivity which will enable the efficient delivery of other infrastructure programs

⁷ In 2005, it was estimated that about 85% of the total population was living in slums or partially destroyed housing. About 40 villages were damaged or destroyed by the tsunami of December 2004. Source: <http://www.encyclopedia.com/topic/Somalia.aspx>

⁸ Somalia IDP Solutions Initiative, Prof. Walter Kaelin, Special Advisor to the DSRSG/RC/HC on Internally Displaced Persons, 'Mission report 27 Jan to 11 Feb 2016'

⁹ Source: <http://worldpopulationreview.com/countries/somalia-population/>

¹⁰ Compare 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015

and other development plans. There is the potential to link Somalia with land-locked countries in the region (Ethiopia, South Sudan, Uganda and Central Africa) through rehabilitation of inter-state and inter-regional road corridors, development of railway & air connectivity and also transit service facilities.

4.3.1 Roads¹¹



Map 1: Overall road network of Somalia¹²

The 21,933km long road network in Somalia is in poor to very poor condition. 90 % of the primary roads have deteriorated and are well past their designed life-span.¹³ With a huge maintenance backlog, there is a significant need for reconstructing or resurfacing these existing paved roads throughout all of Somalia. Within the country, only 2,860 km (13%) of a total 21,830 km of roads are paved. The rest of the network is earthen or gravel. A major part of the investment in the roads and transport program can deliver livelihoods and cash transfer benefits while improving the infrastructure. Overall in Somalia, approximately 80% of the population living in settlements, towns and cities resides in only 10 large cities¹⁴, this is 50 per cent of the total Somali population. In the short term, priority also needs to be given to roads serving main cities, towns and settlements. This will help create employment, link markets, and reduce the cost of doing business, making Somali enterprises more competitive. Somalia is also not well connected to its neighbors. This limits economic activities such as trade within the region. An independent Roads Agency on the federal level does not exist for the time being but would significantly contribute to coordination of rehabilitation and maintenance efforts in the country.

¹¹ 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015, p. 68 ff.

¹² 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015

¹³ Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015, p. 68 ff.

¹⁴ Mogadishu, Hargeisa, Bosasso, Galkayo, Berbera, Merca, Jamame, Borama, Kismayo and Baidoa

Competences for road construction, rehabilitation and maintenance are so far distributed between several ministries decreasing the efficiency of all ongoing and intended interventions.

Road classification and surfacing data:¹⁵

Type of surfacing	Primary/ Main roads	Secondary roads	Rural/Feeder roads	All roads	Percentage
Paved	2,442	418	-	2,860	13%
Gravel	-	844	-	844	4%
Earth	220	3,588	14,421	18,229	83%
	2,662	4,850	14,421	21,933	100%

4.3.2 Ports¹⁶



Map 2: Existing Somali ports¹⁷

Somalia's strategic location at the Horn of Africa on the Suez Canal shipping route and the longest country coastline (3,330 km) in Africa offers a great advantage to port operators. Despite the long shoreline Somalia has only four deep-water ports. Built in the 1980s, these four major operational ports in Somalia are situated in the economic centers of Mogadishu, Berbera, Kismayo and Bosaso, with only Mogadishu and Berbera handling significant volumes of traffic.

¹⁵ Infrastructure Cluster Report, World Bank, 2006

¹⁶ 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015, p. 98 ff.

¹⁷ 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015

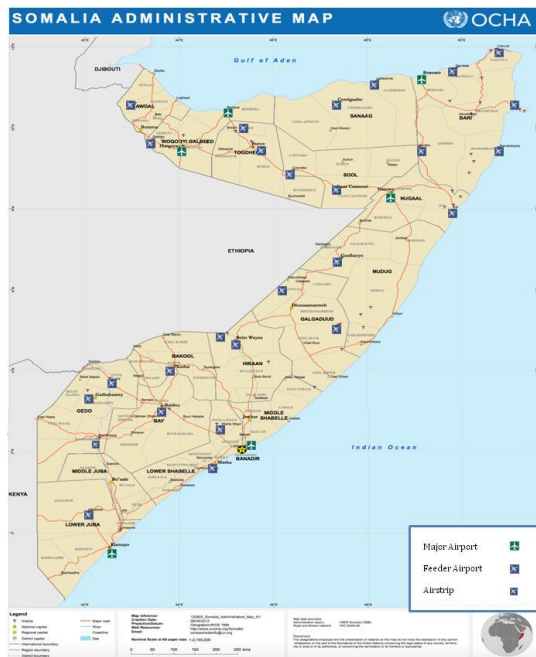
All these ports were originally under the control of the Somali Ports Authority, which no longer exists.

Somalia's ports are important for the country's exports, which are heavily concentrated around livestock that makes up 81% of all the total exports, oil seeds (10.5%) and animal hides and skins (2.4 %) with Saudi Arabia being the largest trade destination accounting for over 55% of all exports by value. Largest import sectors are the construction sector and related industries as well as food items (vegetables, flour, cereals, dairy products and tobacco), together comprising 62% of total imports. Though there is easing of security situation, Somali ports had been badly affected by piracy with 237 cases reported in 2010, 75 in 2012 and 15 in 2013.

Despite the devastating effects of the civil war and due to rehabilitation activities undertaken in support to humanitarian operations between 2007 and 2013, these major ports are in reasonably good condition. However, the ports face considerable challenges in augmenting their operating performance, which is poor for two main reasons: First, the ports have limited equipment, with most of the cargo handling being carried out using the gear of the ships themselves. Second, they are operated by the government with a large pool of unionized dockworkers and based on outdated working practices. These challenges limit the capacity of the port to meet the demands of economic growth. As the primary mode of importing and exporting economic goods into the country, the capacity of ports needs to be enhanced in order to strengthen the potential for economic growth. To facilitate exports rehabilitation of feeder and coastal roads linking the ports to interior agricultural productive zones of the country are needed.

4.3.3 Airports¹⁸

¹⁸ 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015, p. 81 ff.



Map3: Major airports and airstrips in Somalia¹⁹

Somalia has an estimated number of 60 airstrips, airfields and airports evenly spread across the country and developed since independence mainly for military strategic reasons. There are 7 major feeder airports, and 23 other important airstrips. The 6 main airports include Mogadishu, Berbera, Hargeisa, Bosasso (recently upgraded), Garowe and Kismayo. Almost all other major urban centers also maintain at least one airstrip that is capable of handling a small aircraft.

With its thin population, spread out over a large area with only a few large urban centers and with a roving pastoral population of 40%, airports are critical for national and international connectivity, access and mobility. Airports and the aviation industry are a critical backbone of the national economy due to its important role in delivering critical humanitarian and emergency assistance to large vulnerable population groups, ensuring high value trade benefits for the economy and as a generator for revenue and job creation. Somalia's per-capita income is among the lowest in the world. As a result, the ability of Somalis to pay for commercial aviation is low. Somalia's aviation sector is underdeveloped in comparison with other African states, as well as with the rest of the world. Somalia has very low levels of commercial aircraft activity and passenger movements, and some inherent limitations with regard to the operating environment, political instability and internal conflicts. The various regions and governments need to have a well-defined role as regulators and facilitators of growth in the aviation environment.

The absence of a Somali Civil Aviation Authority is a major challenge to control and develop of the aviation sector, as well as the lack of aviation equipment and skilled human resources. Adan

¹⁹ 'Transport Sector Needs Assessment', FGS, AfDB, EU and UNOPS, Dec 2015

Abdulle Osman Mogadishu airport is almost unable to handle the present traffic demand in and around Mogadishu. In a short while the demand can double in comparison of the present situation. Mogadishu definitely needs a new airport that can handle the increasing demand of air traffic.

4.4 Development and Management of Water resources and Waste infrastructure

4.4.1 Water

A large portion of Somalia's population relies on uncontrolled, low quality shallow-wells for its water supply. Majority of the water sources in rural areas are dug wells (62%) and boreholes (19%) complemented by surface water sources. Urban areas depend mainly on deep groundwater complemented by dug wells for water supply. Overall access to improved water supply in Somalia increased from 21% in 2000 to 30% in 2011. However, proportion of rural dwellers with access to water and sanitation significantly decreased from 15% to 7%, while that of urban areas increased from 35% to 66% in the same period. Existing urban and rural infrastructure systems are aged and in some cases destroyed during the prolonged civil war. About 60% of existing water supply infrastructure assets require rehabilitation, expansion or replacement in both urban and rural areas to meet the increasing demand.

The extreme shortage of surface water in parts of the country, especially in Puntland and Galmudug states, means that groundwater is the only reliable water source and this is at depths of up to 400 meters in some areas. Due to the depth of the groundwater, rural water supplies often rely on diesel-powered pumping systems, which require regular maintenance from skilled mechanics, who are only found in urban areas. Studies show that up to 40 per cent of the water supplies are non-functional at any one time. In urban settings where water systems have been established in the past they are usually operated by private or public-private partnerships. These urban water supply services are often uncoordinated and unregulated resulting in inefficiency and non-optimal use of available resources. In other urban areas surface water or shallow wells are used, but are at high risk of contamination, as most urban waste is disposed of into open drains.

4.4.2. Waste Management and Sanitation infrastructure

The water supply sector is confronted with the challenges to accelerate delivery of improved water infrastructure services while lacking institutional and organizational capacities, policies and regulatory framework, adequate sector financing, and skilled labor, that would be key to building the capacity of the sector, enhancing its operation, maintenance and delivery mechanisms. Absence of adequate infrastructure for harnessing the available surface and ground water potential hinders easy, reliable and equitable access to water supply for domestic and agricultural uses.

In urban areas, while some pilot initiatives have explored sustainable approaches towards solid waste management, most are characterized by informal dump sites, dirty streets and absence of sustainable solid waste collection systems and facilities. In Somalia, municipalities have limited human and technical capacity and skills to carry out sustainable and systematic solid waste collection and management. In some cities small private companies are involved in waste collection and disposal but contracts with municipalities are often of short duration, a fact that hinders larger investment in technical equipment. Urban communities are used to disposing off household, business and hospital waste on the streets or informal dump sites and are reluctant to pay for regular waste collection and management. Urban residents are, after more than 20 years of conflict, used to irregular city clean up campaigns based on cash for work type activities and lack understanding of the importance of sustainable and regular solid waste collection and management. Urban poor, in particular women and youth, often collect household and business waste for a small fee and dispose it unregulated and uncontrolled. The few existing formal waste disposal sites do not meet minimum environmental standards. There is a huge potential for private-public partnerships centered on recycling and reuse which can generate employment and SME opportunities, biomass energy, natural fertilisers, etc. In rural areas there are no solid or liquid waste disposal methods. Knowledge of the risks to health is very poor and there is no incentive or motivation for people to dispose of their waste properly.

Sanitation

Access to improved sanitation in urban areas jumped from 45% in 2000 to 52% in 2014. About 3% of the urban dwellers resort to open defecation. However without any functioning sanitation system or sewers in the urban areas, most human waste is disposed of directly into open drains. Access to improved sanitation in rural areas decreased from 10% to 6% over the same period. Approximately 83% of the rural population practices open defecation. This explains the observed high incidence of water and sanitation related diseases. . For the past 25 years the incidence of diarrhea in children under five years has remained at alarming high level of 23% in spite of improvement in access to improved water sources. About 40% of sanitation infrastructure systems require rehabilitation in both urban and rural areas

There are no sewerage systems in urban areas of Somalia and in rural areas less than one third of households have their own latrine. Most schools and health centres have very poor water and sanitation infrastructure. Where these exist there are no recurring funds to ensure maintenance and repair.

4.5 Energy²⁰

The limited availability, reliability and the high cost of energy are major constraints on economic growth in Somalia. The energy mix in Somalia is reliant upon locally available charcoal and

²⁰ Compare 'Energy Sector Needs Assessment' , FGS and AfDB, Aug 2015

firewood as the core sources of energy, meeting nearly 80 to 90% of the energy needs of the whole country and resulting in excessive exploitation of these primary sources. Annual consumption of charcoal is estimated at around 4 million tons per year, equivalent to about 400 kg/capita/annum, a rate that is fast exhausting Somali's few forests. The prevalence of charcoal and wood for cooking also has serious health impacts at the household level that can be mitigated by the introduction of cleaner, more efficient and cost-effective end-use devices and modern and environmentally friendly cooking fuels. Petroleum products, essentially for transport, electricity generation, cooking and lighting account for about 10% of the total energy use. Electric power generation (almost entirely diesel-fueled) accounts for about 2 of these 10%. Several cities (Garowe, Hargeisa, Bosaaso, Berbera, Qardo) have small, generally dilapidated grids which are not always in use. Overall, the total installed and operational generating capacity in Somalia is estimated to be 80 to 85 MW at estimated 250,000 connections. Drops in voltage are frequent and supply is limited to 5 to 6 hours per day. The low tension (480/220 V) over sometimes long distances leads to tension and technical losses with most small generators reporting 40% to 50% power loss.²¹

On the other hand, renewable energy potential is abundant. The solar energy potential ranges from 5 to 7 kWh/m²/day with over 310 sunny days in a year or 3,000 hours of sunshine per annum. Somalia is also characterized by strong wind regimes with annual average speeds of 1.5 to 11.4 m/s with potential for small hydro power (around 100 to 120 MW) along the Shabelle and Juba rivers.²²

Due to the lack of investment, the volatile security situation and the absence of public resources and public oversight, the potential to cultivate electricity and expand electricity infrastructure systems and markets has been hampered over the past two decades. Renewable and alternative energy sources, such as solar and wind power, present substantive opportunities to diversify and expand the energy infrastructure systems of Somalia but so far has not been harnessed, due to both security and funding problems. Developing both fossil and renewable resources will require a functioning transport and logistics sector and fiscal regimes which promote private sector investments. Major challenges experienced within the energy sector include inadequate qualified and skilled personnel as well as limited access to, generation, transmission and distribution of electricity power, depletion of biomass, and low penetration of modern and environmentally friendly energy, especially in the rural areas.

Some investments have been made in solar energy by international donors and aid agencies. In the water and health sectors there are successful solar powered pumping systems and vaccine refrigeration units working in health centers across the country, sustainably for almost a decade.

4.6 Information Communication and Technology (ICT)²³

²¹ 'Energy Sector Needs Assessment', FGS and AfDB, Aug 2015, p.11

²² 'Energy Sector Needs Assessment', FGS and AfDB, Aug 2015, p.15

²³ Compare 'Needs Assessment of ICT Sector in Somalia', FGS and AfDB, July 2014

The ICT sector in Somalia has flourished under a self-regulated private sector regime and is known to be a key contributor to the economy. Somalia's ICT market consists of 11 operators and 4 million mobile connections, but tele-density is only about 7% and the proportion of Internet users is growing rapidly. However, ICT is one of the fastest growing sectors and the third largest industry by employment in the country. The sector employs around 25,000 Somalis, generates substantial profit and has enabled a mobile penetration rate of about 10 subscriptions per 100 people. Most readily available ICT media are radio and mobile phones. The TV and press sectors are weak, with radio being the dominant medium. The country has more than 50 FM radio stations. Mogadishu alone has 20 stations.

86% of households have access to radio and 26% to TV. Radio and TV sub-sectors need to be further developed as employment and revenue generating sectors.

However, Somalia has just developed ICT policy and regulations to guide infrastructure roll-out as well as the development of ICT applications such as mobile money transfer. The country's volatile security and lack of stable energy sources greatly hinder growth in the ICT sector. Low literacy and ICT skill levels among vast groups of the population limit demand for ICT services in the country.

Somalia has access to the East Africa Sub-marine Cable System (internet) with Mogadishu and Bosaaso so far being the only landing points that also connects to local optic-fibre cable lines. Somalia is, however, extremely close to several major undersea cable networks which connect Europe, the Gulf Region in Middle East and India through the Gulf of Aden and transfer a major share of the globalized Information Economy of the 21st century, for example through the Europe-India-Gateway (EIG) or India-Middle East-Western Europe (I-ME-WE) network. The strategic position of Somalia has not been explored when it comes to investment opportunities for the growing number of educated young people who do not find adequate jobs at the domestic labor market.

5. The Main Challenges hindering the Infrastructure Sector growth and expansion

Despite the progress made towards peace and stability, Somalia continues to face major challenges in revitalizing the infrastructure sector. Systemic challenges include weak institutional frameworks and the absence of an integrated and long term national infrastructure plan that is grounded in a sustainable approach to development for all the regions in the country.

The absence of and lack of funding to enforce harmonized policies and regulatory frameworks at the federal and local levels of government including basic standards such as building codes also inhibits the development of the sector as it prevents the capacity of government structures to effectively regulate ongoing projects. The lack of water quality standards hinders the possibility of regulation of private water supply companies, and the control of industries which might cause pollution to water sources. This challenge is further exacerbated by a dearth of infrastructure professionals and a skilled labor force, who can substantively contribute their technical capacity to the development of the sector. The absence of both is adversely impacting the capacity to adequately deliver, monitor and maintain quality infrastructure systems.

Dual claims over land, unresolved land disputes and informal occupation of public land hinder both private and public investments for infrastructure assets, in particular in the urban areas.

The extremely limited capacity of Somalia's existing four deep-sea ports combined with the high percentage of imported construction materials (hardly any in-country production) does not only lead to high construction and transportation costs but also creates a serious bottle neck for scaling up construction and rehabilitation investments and efforts.

Gaping budgetary constraints and Somalia's inability to access international financing for large-scale infrastructure projects have also limited the possibilities of investment in the sector. Because of the lack of a centralized coordination and regulatory mechanism, haphazard sector development activities hamper the realization of the long term national vision, underpinned and delivered through a systems-based approach to long term national infrastructure planning. Furthermore, there are no significant budget allocations for the maintenance of infrastructure investments.

The volatile security situation in large parts of the country poses specific security threats that hinder expansion of the infrastructure sector: ensuring security of workers, materials and works during rehabilitation works in insecure areas, safeguarding long infrastructure networks/pipelines after completion, creation of critical but vulnerable infrastructure assets in particular with regards to energy plants and securing large scale immobile investments.

As security challenges continue, investment in the development of private and public infrastructure will struggle to take off.

6. Infrastructure Sector's *Development Objectives*

Overcoming the challenges to infrastructure rehabilitation, social service provision and human capacity to plan, operate and maintain the infrastructure assets as well as connecting investment in infrastructure to the long-term development of Somalia is the overarching long-term objective of the infrastructure development sector. At the same time there is need to put in place short to medium term practical and cost effective infrastructure sub-sector programs that consider poverty reduction (SDG1), build resilient infrastructure (SDG9), ensure access to affordable energy (SDG7), achieve equitable access to safe and affordable drinking water and sanitation (SDG6) and make cities safe, resilient and sustainable (SDG11), improve food and nutrition security through the use of more reliable and accessible cooking fuels (SDG2) and sustainably manage forest, combat desertification, adapt and mitigate against climate change (SDG13) and protect biodiversity through substitution of biomass (SDG15). Through an integrated approach to infrastructure system development, stability as well as economic, socio-political and environmental sustainability will be achieved.

Overall Objective for the infrastructure sector in 2018 to 2023:

Revitalize, operate and maintain essential infrastructure and services to improve stability and set foundations for social, economic and sustainable development.

Key among the infrastructure sector's development objectives include:

- i. Strengthen policy and regulatory frameworks at the Federal Infrastructure Ministries to guide, plan and implement infrastructure development and related social service provision;
- ii. Strengthen institutional and technical capacities of different infrastructure sub-sectors to be able to plan, implement, operate, maintain and monitor sector specific infrastructure and service development;
- iii. Sub-sector specific objectives include:
 - a. Improve mobility and connectivity throughout Somalia and to the neighboring region and the world with a strong focus on rehabilitation and maintenance of existing infrastructure (such Airports, SeaPorts and Roads) while protecting the Environment and cultural heritages;
 - b. Strengthen urban and territorial planning processes,
 - c. Improve access to clean energy systems and equitable access to safe and affordable drinking water and sanitation,

- d. Increase access to adequate and affordable housing in particular for IDPs, returnees, the urban poor and other persons in vulnerable situations,
- e. Provide telecommunication networks that ensure free and rapid flow of information.

7. The Infrastructure Sector *Strategic Policies*

Development of several sector specific policies will provide a strong legal framework to support stability and infrastructure system development in Somalia. Key among the policies needed to facilitate this includes:

7.1 National Urban/ Land Policy

Land related conflict is by far the largest driver of conflict across Somalia.²⁴ Lack of clear land tenure system with customary and formal land ownership systems operating together affects not only infrastructure development but also reintegration of the Internally Displaced Persons (IDPs). Destruction of land registries, related information and lack of a functional legal system makes land administration even harder. Unresolved occupation of land and property requires sustainable solutions to enable medium and large-scale infrastructure investments, proper urban planning and sustainable housing solutions. A functioning system of land administration needs to be re-built. A land policy that creates institutions and laws to meet claims for property restitution and that shapes future social and economic structures is needed, without which infrastructure and other investment will be deterred. Urban land policies and related legal and regulatory frameworks would prevent overcrowding, public health risks and the development of slums. As a first step towards a practical and functional solution based on successful local interventions and best practices needs to be developed to accommodate the instant need for urban planning and serviced land.

7.2 National Housing Policy

Housing rights challenges are common in Somalia like in other post conflict countries. Post conflict housing reconstruction would play an important role in establishing the Somali's development and prospect of peace. A National Housing Policy addressing widespread property destruction, mass population return and internal displacement, reconstruction needs of the post conflict communities, innovative housing finance mechanisms and promotion of local building culture is needed.

7.3 National Transport Policies

Policies and regulatory frameworks for the transport sector are required to ensure development of effective, efficient, and safe land, air and marine transport systems. Regulations need to steer

24

both public and private investments in line with the national priorities and support economic growth through the provision of better access to employment and enhance social service provision.

Establishment of competent transport authorities or administrations is essential for management of transport sector resources. Preparation and ratification of the new codes as well as other relevant international conventions is mandatory to regulate the transport sector operations and give confidence to potential investors.

7.4 Sustainable Energy Investment Policy and Energy Strategy

The Federal Government of Somalia does not have any policies or regulations specific to the energy sector. The Energy Sector Management Department of the Federal Ministry also lacks human and financial capacity. Little capacity is available to develop legal and regulatory frameworks; hence the sector operates in a vacuum. However, Somalia is endowed with renewable energy resources to increase affordable energy supply for Somalia. To harness the potential of renewable energy sources, strengthen the energy sector, increase its efficiency and affordability, policies, related regulations and strategies for implementation are inevitable. As a first step a Sustainable Energy Investment Policy focusing on energy efficiency, renewable energy and promoting decentralized renewable energy systems as well as the development of an Energy Strategy for the whole country, which takes into account the role of the private sector, are essential.

7.5 Waste Management and Sanitation infrastructure Policy

The water supply, sanitation infrastructure and waste management sector in Somalia is generally driven by a vibrant private sector with complementary services from the public sector, primarily in urban areas. Despite the public sector playing a facilitative role, the sector lacks a policy and regulative framework hindering adequate and systematic sector development. Urgent institutional strengthening and sector capacity building to address institutional and organizational development and capacity deficiency challenges is needed. The Inter-Ministerial WASH Steering Committee is currently working on the development of a WASH Policy. This policy will cover all aspects of WASH in Somalia including water supply and quality, sanitation and waste management, hygiene education and behavioral change. The WASH Policy will have a human rights-based approach and is aligned to the relevant SDGs. The policy will define roles and responsibilities for this overlapping sector.

7.6 ICT Policy

The area of the policy and regulatory implementation is the number one priority for development of Somalia's ICT sector to harness the potential of further investments, job creation and expanded post and e-service provision.

8. The Infrastructure Sector *Development Priorities*

Infrastructure development in Somalia currently takes place on land only. Capacity and institutional development in land and urban policy development including reconstitution of land registries provides a foundation for land-based infrastructure system development. Somali cities are growing quickly, as a first step a simple and basic planning framework is needed to plan for urban growth and identify and plan city extension areas to accommodate and integrate IDPs and refugee returnees and provide housing strategies for the urban poor and rural - urban migrants.

Somalia's **road networks** are nearly all earthen. Those paved are mostly in poor to very poor condition. Maintenance and the need for reconstruction or resurfacing of existing paved roads need to take priority, whilst also taking into consideration the need to ensure their resilience to climate change and variability. However, this needs to go hand in hand with institutional and technical capacity development and with a robust maintenance regime that requires regular funding from the annual budget. There is further a need for the establishment of road funds and innovative ways of generating for regular road maintenance.

A safe, secure and cost-effective aviation regulatory system will underpin the future development of air services and instill confidence to operate and invest. Somalia needs to establish a credible regulatory platform from which to develop, service, and fill gaps in the current aviation system. Priority should be given to establish the Somali Civil Aviation Authority (SCAA). The other priority is the construction of the new gateway Mogadishu airport which can handle the increasing demand of the air traffic in the next decennia. Also, the re-establishment of the national carrier is a priority for the Somali Civil Aviation ministry.

Somalia port's tonnage is expected to increase to around 3.3 million tons by 2025, an almost 65% increase. Given the present berth occupancy rates of 20% at below par handling speeds and high costs, improvement in specialized handling capacity, hinterland connectivity and port information system to track its performance is of great priority.

The improvement and increased access to energy through use of renewable energy and to substitute biomass fuels in households is vital. Actual investments should focus on providing affordable, consistent and renewable energy at scale. However, technical capacity development for the sector authorities is needed to roll-out basic regulatory framework to guide development.

The area of policy and regulatory implementation is the number one priority for development of Somalia's ICT sector. This includes the development of a framework to implement the telecommunications sector policy and regulations to build Somali ICT infrastructure. Independent regulatory authority is established and key regulatory issues are identified which is great importance for Somali ICT infrastructure investment. Furthermore, there is also need to

ensure that Somalia complies with the international cut-off date for migration from analogue to digital transmission.

In short to medium term, strategic interventions in water and sanitation will be rehabilitation and/or extension of rural and urban water and sanitation infrastructure systems to improve service delivery and ensure sustainable management. Institutional strengthening and sector capacity building to address institutional and organizational development and capacity deficiency challenges is needed as well as ground water resource mapping to provide updated and more reliable data on aquifers, including aquifer monitoring and management to avoid over exploitation.

9. The Main Infrastructure Sector's *Programs during the Plan Period 2019-2023*

i. Institutional Capacity Building Program for the Infrastructure Ministries

- Support in development of policies and regulatory frameworks governing the sector and linking them to those of international and regional associations and bodies such as EAC, COMESA, NEPAD, ICAO, IMO etc.,
- Undertake comprehensive capacity enhancement, including development of standards, procedures and technical trainings on effective reconstruction, infrastructure financing, assessment and planning, managing, maintenance and procurement of public infrastructure works in Somalia.

ii. National Land, Physical Planning and Urban Solutions Program

- Develop national land and urban policies with a focus on climate change adaptation and mitigation and natural disaster planning,
- Lay-out foundation or road map for reconstruction of national land cadaster and registry,
- Develop national public land inventory – to facilitate infrastructure system development,
- Strengthen institutional and technical capacities (human resource and technical skills) of national and regional planning institutions, including in economic, social and environmental assessments (e.g. Strategic Environment Assessments),
- Facilitate planning of regional development corridors and settlement systems,
- Develop regional and urban strategic plans with special focus on urbanization and local economic development, and develop spatial strategic plans for all state capitals and other urban centers which guide city extension and other urban development plans,

- Undertake urban renewal/ reconstruction and expansion including rehabilitation of public space,
- Develop and implement Urban Regulatory Frameworks for new states and for Mogadishu.

iii. National Housing Program: Adequate, Affordable and Sustainable Housing

- Develop resettlement and integration solutions that are durable and at the same time upgrade conditions of temporary IDP shelter to meet minimum living standards through participatory pilot projects,
- Develop National Urban Housing Sector Profile and National Housing Policy,
- Facilitate access to land for housing development,
- Identify and promote suitable and sustainable alternative construction techniques, and promote a local building culture, which makes use of local raw materials and labor, is affordable and of low embodied energy, and spurs local economic development in the short and long term,
- Identify and, if necessary, establish supply chains,
- Establish municipal housing programs (including rental subsidy programs or owner driven incremental reconstruction using micro-credits for purchase of building materials).

iv. Roads Improvement Program

- Develop policies and regulatory frameworks to guide future roads maintenance, expansion and related service delivery,
- Establish an independent Highway authority at federal level,
- Develop a robust maintenance regime,
- Establish reliable and regular funding for the road agency in order to ensure no medium-term collapse of the existing road system,
- Strengthen institutional and technical capacity to design, build and maintain roads,
- Plan, rehabilitate and expand inter-state and inter-regional road corridors,
- Plan, rehabilitate and expand urban roads,
- Create employment opportunities through labor-intensive approaches to roads rehabilitation incorporating appropriate labor policies, standards and procedures,
- Prioritize use of locally produced construction materials to enhance private sector development and employment creation.

v. Ports (Upgrading) Program

- Establish Somali Maritime Administration/Authority,

- Develop detailed Somalia Ports Development Strategy, update Somali Merchant Shipping Act, prepare and ratify the new maritime code and localize International Conventions for shipping and seafarers,
- Institutional strengthening of authorities responsible for managing different ports,
- Expand/ upgrade the existing deep-sea ports and develop strategy for one new port, incl. rehabilitation of seamarks and lighthouses of the Somali coastline,
- Rehabilitate and expand existing cost barge berths,
- Harmonize port charges as a source of government revenue and facilitate sustainable use by local exporters,
- Conduct feasibility studies and technical assessments for planning future investments,
- Extend network of secondary ports and jetties to facilitate development of local fisheries and improve sea-based transport links.

vi. Aviation Infrastructure Program

- Reform Aviation Sector Policies and relevant Acts,
- Carry out infrastructure rehabilitation at the main regional state airports, including rehabilitation of terminals,
- Conduct feasibility study for a new international airport in Mogadishu,
- Develop capacity and capability of regulatory and operators personnel,
- Improve basic air navigation services and reconstruct the airport infrastructure to provide safe passenger and freight services to/from/via Somalia.
- Establish the Somali Civil Aviation Authority (SCAA).
- Finalize the activity concerning the re-establishment of the national carrier (Somali Airlines)

vii. Energy and Water resources Service Expansion and Affordability Program

- Rehabilitate vital state infrastructure such as key public institutions to enhance social service provision,
- Establish hydro-meteorological Services institution under the ministry to observe and monitor weather, climate and River water levels and meet the needs of end uses.
- Develop sustainable energy investment, water and sanitation policies, and national energy provision strategy,
- Develop institutional capacity to regulate, operate and maintain energy, water and sanitation utility infrastructure and service provision,
- Rehabilitate and expand energy, water and sanitation systems,
- Expand access to clean energy (solar, wind, biogas) and reduce biomass based energy sources like fuel wood and charcoal,

- Undertake assessment and map renewable energy resources, incl. wind pattern maps to harness wind energy, and erect wind masts,
- Map ground water resources to provide updated and more reliable data on aquifers, including aquifer delineation, monitoring and management to avoid over exploitation,
- Support the development and implementation of Integrated Solid Waste Management Plans in cities and promote sustainable and regular solid waste management, develop respective legislation and implement pilot projects at municipal level.

viii. ICT Program

- Implement the developed ICT Policy and Strategy for Somalia that would incorporate a national broadband plan and an e-Government plan,
- Operationalize the established independent ICT regulatory agency (National Communication Authority),
- Establish the national backbone network, to connect the major cities, and federal member states, including through public private partnership (PPP) as appropriate,
- Issue licenses for infrastructure-based telecommunication operators, and spectrum permits for mobile operators, with appropriate obligations, including taxation of telecom services,
- Apply market mechanisms for spectrum management, including introducing a licensing process for 4G and 5G spectrum and a strategy for migration from analogue to digital broadcasting,
- Introduce SIM card registration for all mobile devices,
- Establish local loop network
- Providing Backbone connectivity to Federal and Regional states
- Establish X number of communication rooms at national and state level
- License Media and Operators
- Re-establish postal service throughout the country
- Expansion of connection to secondary schools, technical institutes and vocational training institutions
- eGovernment strategy and implementation to increase government revenues and facilitate service delivery
- Establishment of integrated tech-based services offices (One-stop shops for public services from IDs, passports to Ministry of Commerce's business registration, etc).

10. Aggregated Resources Requirement

(see table below on infrastructure programs)

11. Milestone of *Infrastructure Sector* development (2019-2023)

- Infrastructure sector institutions capacities are strengthened (skilled human capacity injected, the necessary sub-institution, structures & legislation established, systems for prioritization of investment developed) and can lead the country's infrastructure development progress,
- Flagship programs in the infrastructure of each sector (e.g. ICT, transport, energy, livestock, agriculture, water & sanitation, housing and vital (state) government infrastructure) are initiated and/or under implementation.
- Infrastructure strategy, master and investment plan is presented and leads to financial inflows to enable implementation and SITF is fully functional, including spatial plan for post NDP development programmes (2020-).

2030

2063

12. Coordination aspects and requirements with other Sectors/activities/regions

Inter-ministerial Infrastructure (Public Works) Coordination Mechanism (IMPWCM) consisting of Ministry of Public Works, Housing & Reconstruction, Ministry of Transport & Civil Aviation, Ministry of Ports & Marine Transport, Ministry of Energy & Water Resources, Ministry of Post & Telecommunication and the Environmental Agency of the Office of the Prime Minister will be the leading entity to guide development of policies, strategies, legislation and implement infrastructure programs. Ministry of Public Works, Reconstruction & Housing is the lead ministry and is chairing the IMPWCM. IMPWCM will soon include regional state members, international development partners and interested investors. IMPWCM will function as a platform where the infrastructure development ideas will be raised, planned and executed. Ministry of Public Works will, in close cooperation with Ministry of Planning and International Cooperation (MOPIC) and Ministry of Finance (MoF) be responsible for the coordination with the other sectors (e.g. productive sector, social sector, agriculture, fishery, industry & tourism), national and international stakeholders, donors and investors to discuss infrastructure development ideas and strategies to develop Somali infrastructure systems.

There is also a need for inter-sectorial integrated infrastructure strategy to enhance economic growth and to increase nationwide economic development. The key ministry to coordinate the nationwide economic development programs and to combine sectorial strategies and program is the Ministry of Planning and International Cooperation (MOPIC). The Ministry of Finance (MoF) is responsible for ensuring that adequate financial resources will be available for the Main Infrastructure Development Programmes outlined above.

13. Conclusion

(to be written once draft has been reviewed by all stakeholders)

14. Intervention Programmes

Pillar 6: Infrastructure Sector Intervention Programmes					
Main Programs	Outcomes	Time Frame	Actors Involved	SDG	Estimated Costs (USD)
1. Institutional Capacity Building Program	1.1 Regulatory framework for quality, reliable, sustainable and resilient infrastructure developed to support economic development and human well-being, with a focus on affordable and equitable access for all	2019-2023	All infrastructure ministries	SDG 9 SDG 11	9M
	1.2 Increased institutional capacities for quality, reliable, sustainable and resilient infrastructure developed to support economic development and human well-being, with a focus on affordable and equitable access for all	2019-2023	All infrastructure ministries	SDG 9 SDG 11	8.5M
2. National Land, Physical Planning and Urban Solutions Program	2.1 Inclusive and sustainable urbanization and capacity for participatory, integrated and human settlement planning and management <u>Key activities under 2.1</u> i) Strengthen institutional and technical capacities (human resource and technical skills) of national and regional planning institutions	2019-2023	Ministry of Public Works, Housing and Reconstruction	SDG 11	17.5M

	ii) Develop national land and urban policies (with inclusion of climate change adaptation and mitigation and natural disaster planning) iii) Develop regional and urban strategic plans with special focus on urbanization and local economic development, and develop spatial strategic plans for all state capitals and other urban centers which guide city extension and other urban development plans: 1. State capitals 2. Mayor economic centers 3. Refugee and returnee receiving cities iv) Develop and implement Urban Regulatory Frameworks for new states and for Mogadishu				
	2.2 Increased land tenure security <u>Key activities under 2.2:</u> i) Road map for national public land inventory – to facilitate infrastructure system development drafted and aligned with ongoing activities in federal states and Mogadishu for state public property inventories ii) Work units established at Ministry of Public Works at federal and state level and provided with capacity building trainings	2019-2023	Ministry of Public Works, Housing and Reconstruction	SDG 11 SDG 16	13M

	iii) Integration of eviction guidelines principles into the codes of conduct and manuals for the Ministry of Public Works, State Governments and Municipalities				
	2.3 Vibrant construction sector utilizing local materials that responds to resilience/climate change issues <u>Key activities under 2.3:</u> i) Plan and undertake urban renewal/ reconstruction and expansion including rehabilitation of public space: 1. State capitals 2. Mayor economic centers 3. Refugee and returnee receiving cities 4. Other urban centers ii) Identify and promote suitable alternative construction techniques and a local building culture, which makes use of local raw materials and labour, is affordable and of low embodied energy, and spurs local economic development in the short and long term	2019-2023	Ministry of Public Works, Housing and Reconstruction	SDG 11 SDG 12 SDG 13	18M
3. National Housing Program	3.1 Improved access to adequate and affordable housing for the people of Somalia, in particular IDPs, refugees and returnees <u>Key activities under 3.1</u> i) Develop resettlement and integration solutions that are durable and at the same time upgrade conditions of temporary IDP	2019-2023	Ministry of Public Works, Housing and Reconstruction	SDG 1 SDG 11 SDG 16	19M

	shelter to meet minimum living standards through participatory pilot projects ii) Develop National Urban Housing Sector Profile and National Housing Policy iii) Upgrade housing units and make them available to IDPs iv) Establish municipal housing programs (including rental subsidy programs or owner driven incremental reconstruction using micro-credits for purchase of building materials)				
	3.2 Improved access to land for housing development i) Formulate clear legislation on land management and administration for public and private land that relate to the constitution currently in place ii) Develop provision for affirmative action regarding land concessions for housing purposes to IDPs IDP returnees and other vulnerable groups	2019-2023	Ministry of Public Works, Housing and Reconstruction	SDG 11	5M
4. Roads Improvement Program	4.1 Policies and regulatory frameworks to guide future roads maintenance, expansion and related service delivery <u>Key activities under 4.1:</u>	2019-2023	Ministry of Transport & Civil Aviation; Ministry of Public Works, Housing and Reconstruction	SDG 8 SDG 9	3.7M

	i) Strengthen institutional and technical capacity to design, build and maintain roads ii) Develop policies and regulatory frameworks to guide future roads maintenance, expansion and related service delivery				
	4.2 Rehabilitated and extended inter-state and inter-regional corridors <u>Key activities under 4.2:</u> i) Undertake road / transport corridors planning: 5M ii) Emergency maintenance program: (filling of pot holes/ resurfacing to make the roads traffic worthy till start of complete construction works) -15M iii) <u>Rehabilitate and extend inter-state and inter-regional corridors</u>: Total USD670M 1. Mogadishu-Afgooye - Baidoa-Beled Hawo corridor : 229M 2. Mogadishu- Balad -Jowhar - Beledweyne-Galkayo Corridor: 153M 3. Mogadishu – Marka - Kismayo – Liboi Corridor (Kenya): 202M 4. Berbera-Addis Ababa Corridor (Ethiopia): 65M 5. Repair of road from Hobyo, Galkayo & Ethiopian border: 8M 6. Completion Galkacayo-Galdogob		Ministry of Public Works, Housing and Reconstruction. Ministry of Transport & Civil Aviation;	SDG 8 SDG 9	693M

	Road, feasibility/repair Galkacayo-Gara'ad Road: 2M 7. JALAM – XARFO section of the Garowe – Galkayo corridor: 6.7M 8. Rehabilitation Hargeisa – Kalabydh Road: 4.5M iii) Create employment opportunities through labour-intensive approaches to roads rehabilitation/construction: 2M iv) Prioritize use of locally produced construction materials to enhance private sector development and employment creation: 1M				
	4.3 Rehabilitate and construct urban roads <u>Key activities under 4.3:</u> i) Rehabilitate/ construct primary roads and bridges ii) Rehabilitate/ construct secondary roads iii) Strengthen institutional capacity of municipalities re road rehabilitation and construction	2019-2023	Municipalities	SDG 8 SDG 9	32.5M
5. Ports Upgrading Program	5.1 Enhanced legislative and institutional framework and capacity facilitates ports operations	2019-2023	Ministry of Ports and Marine Transport	SDG 9 SDG 14	6M

	<u>Key activities under 5.1:</u> i) Establish Somali Maritime Administration/Authority ii) Conduct feasibility studies and technical assessments for planning future investments iii) Harmonize port charges as a source of government revenue and facilitate sustainable use by local exporters iv) Update Somali Merchant Shipping Act, prepare and ratify the new maritime code and localize International Conventions for shipping and seafarers				
	5.2 Bottleneck of limited capacity of existing deep-sea ports removed <u>Key activities under 5.2:</u> i) Institutional strengthening of authorities responsible for managing different ports ii) Expand/ upgrade the existing deep-sea ports and develop strategy for one new port, incl. rehabilitation of seamarks and lighthouses; 1. Upgrade/ expand Mogadishu port 2. Upgrade/ expand Kismayo port iii) Develop detailed Somalia Ports Development Strategy, update Somali Merchant Shipping Act, prepare and	2019-2023	Ministry of Ports and Marine Transport	SDG 8 SDG 9 SDG 14	30M

	ratify the new maritime code and localize International Conventions for shipping and seafarers iv) Rehabilitate and expand existing cost barge berths v) Rehabilitation of lighthouses of Somali coastline				
6. Aviation Infrastructure Program	6.1 Enhanced legislative and institutional framework and capacity facilitates airport operations <u>Key activities under 6.1:</u> i) Reform Aviation Sector Policies and relevant Acts ii) Develop capacity and capability of regulatory and operators personnel	2019-2023	Ministry of Transport & Civil Aviation	SDG 8 SDG 9	10M
	6.2 Increased operational capacity of airports, improved operations and security <u>Key activities under 6.2:</u> i) Rehabilitate 8 main regional state airports, including rehabilitation of terminals and runways: 9M ii) Construction of the Mogadishu airport taxiway and rehabilitation of the runway iii) Conduct feasibility study for a	2019-2023	Ministry of Transport & Civil Aviation	SDG 9	22M

	new international airport in Mogadishu iv) Improve basic air navigation services and reconstruct the airport infrastructure to provide safe passenger and freight services to/from/via Somalia				
7. Energy and Water resources Service Expansion and Affordability Program	7.1 Equitable access to safe and affordable drinking water <u>Key activities under 7.1:</u> i) Develop institutional capacity to regulate, operate and maintain the water sub-sectors utility infrastructure ii) Develop water policies and regulations iii) Rehabilitate and expand water systems: 1. Establish a regulatory framework to guide public-private-partnerships in water sub-sector, including clarification on roles and responsibilities, 2. Awareness campaigns to achieve behavioral change, 3. Municipal programs targeting the household level, iv) Establish hydro-meteorological	2019-2023	Ministry of Public Works, Housing and Reconstruction ; Ministry of Energy & Water Resources; Ministry of Health; municipalities; private sector	SDG6	62M

	Services institution under the ministry to observe and monitor weather, climate and River water levels and meet the needs of end uses. v) Map ground water resources to provide updated and more reliable data on aquifers, including aquifer monitoring and management to avoid over exploitation vi) Stimulate and promote groundwater resources exploration and development of deep aquifers for pastoralist and agro-pastoralists besides service water, rainwater harvesting as a supplement.				
	7.2 Increased access to adequate and equitable sanitation <u>Key activities under 7.2:</u> i) Develop institutional capacity to regulate, operate and maintain the sanitation sub-sectors utility infrastructure ii) Develop sanitation policies, regulations and strategies iii) Rehabilitate and expand sanitation systems: 1. Establish a regulatory framework to	2019-2023	Ministry of Public Works, Housing and Reconstruction and line ministries; Ministry of Energy & Water Resources; Ministry of Health; municipalities; private sector	SDG6 SDG 11	27M

	guide public-private-partnerships in sanitation sub-sector, including clarification on roles and responsibilities 2. Awareness campaigns to achieve behavioral change, 3. Municipal programs targeting the household level iv) Support the development and implementation of Integrated Solid Waste Management Plans in cities and promote sustainable and regular solid waste management based on user fees, develop respective legislation and implement pilot projects targeting bio-medical waste, recycling, waste to energy transformation and public awareness raising campaigns. Scale up existing best practices from previous pilot projects and provide solid infrastructure systems for environmental friendly waste disposal and recycling.				
	7.3 Ensure access to affordable, reliable, sustainable and modern energy for all <u>Key activities under 7.3:</u> i) Develop institutional capacity to regulate, operate and maintain the energy sub-sectors utility infrastructure ii) Develop sustainable energy investment policies and national energy provision strategy iii) Expand access to clean energy (solar,	2019-2023	Ministry of Energy & Water Resources; Ministry of Public Works, Reconstruction & Housing and line ministries; municipalities; private sector	SDG 7 SDG 9 SDG12 SDG13	79M

	wind, biogas) and reduce biomass based energy sources like fuel wood and charcoal: 1. Increase efficiency of cooking systems, 2. Extend renewable energy products and services for lighting and cooking, 3. Vocational training programs for clean/green energy, 4. Municipal programs targeting the household level, 5. Larger scale infrastructure investments for solar, wind and biogas plants, etc. (post 2019) iv) Rehabilitate and expand energy systems: 1. Awareness campaigns to achieve behavioral change, 2. Promote decentralized energy systems, in particular mini grids 3. Municipal programs targeting the household level v) Undertake assessment and map renewable energy resources, incl. wind pattern maps to harness wind energy				
	7.4 Build quality, reliable, sustainable and resilient infrastructure to support economic development and human well-being, with a focus on affordable and equitable service provision <u>Key activities under 7.4:</u> i) Establish a regulatory framework to guide	2019-2023	Ministry of Public Works, Housing and Reconstruction and line ministries;	SDG11 SDG16	12.5M

	public-private-partnerships for utility service provision including clarification on roles and responsibilities ii) Rehabilitate vital state infrastructure/ minimum 3 key public institutions to enhance social service provision: 7.5M				
8. ICT Sector Program	8.1 Significantly increased access to information and communications technology <u>Key activities under 8.1:</u> i) Implement the developed ICT Policy and Strategy for Somalia that would incorporate a national broadband plan and an e-Government plan, ii) Operationalize the established of an independent ICT regulatory authority (National Communications Authority), iii) Extend the national backbone network, to connect the major cities, and federal member states, including through public private partnership (PPP) as appropriate, iv) Issue licenses for infrastructure-based telecommunication operators, and spectrum permits for mobile operators, with appropriate obligations, including taxation of telecom services,	2019-2023	Ministry of Post, Telecommunications & Technology	SDG 9	70M???

	v) Apply market mechanisms for spectrum management, including introducing a licensing process for 4G and 5G spectrum and a strategy for migration from analogue to digital broadcasting, vi) Introduce SIM card registration for all mobile devices, vii) Establish local loop network viii) Providing Backbone connectivity to Federal and Regional states ix) Establish X number communication rooms at national and state level x) License Media and Operators xi) Expansion of connection to secondary schools, technical institutes, vocational training institutions and universities.				
--	--	--	--	--	--